

## ⚙️ Social Sandbox: Core Functionality & Strategic Importance

This document breaks down the technical operations of the Social Sandbox and explains *why* each feature exists. This is your ultimate cheat sheet for your hackathon pitch: it bridges the gap between the code you wrote and the real-world problem you are solving.

### 📊 1. The Lounge (Asynchronous Dashboard)

#### Functionality:

- **Profile Management:** Users select a persona (e.g., "Job Seeker", "Awkward Texter") which initializes their MongoDB document.
- **Persistent State:** Tracks the user's XP as they pass/fail modules.
- **Global Leaderboard:** Fetches and displays top scores and recent activity from other users in the MongoDB Atlas database.

#### The Importance:

- **Real-World Impact:** Gamification keeps users engaged. By seeing others practice, it normalizes the idea that social skills are a muscle that can be trained, reducing stigma.
- **Hackathon Strategy:** This is your direct ticket to the

*MLH*

Best Use of MongoDB Atlas prize. It proves your app isn't just a static page, but a data-driven platform with persistent cloud storage.

### 💬 2. Sarcasm Sweeper (Texting Simulator)

#### Functionality:

- **Dynamic Generation/Rendering:** Displays an iMessage-style group chat.
- **Interactive State:** Users use Vue-bound checkboxes to "sweep" and identify texts with hidden subtext (e.g., passive-aggression, sarcasm, genuine anger).
- **AI Debugger:** Sends the selected texts to the Vultr-hosted Llama 3 instance, which returns a structured JSON evaluation (accuracy score and psychological breakdown).

#### The Importance:

- **Real-World Impact:** Digital miscommunication is a massive source of anxiety for Gen Z. This trains users to read "between the lines" of text messages in a safe, low-stakes environment.
- **Hackathon Strategy:** Shows you can use an LLM as an *analytical engine*, not just a chatbot. Wrapping the LLM output in a clean, structured UI (The Social Debugger) hits the **Design** and **Technology** judging criteria hard.

### 3. Tone Check (Voice Simulator)

#### Functionality:

- **Speech-to-Text (STT):** Uses the browser's native Web Speech API to transcribe the user's spoken response to a difficult scenario (e.g., setting a boundary with a boss).
- **AI Processing:** Routes the transcript through Flask to Llama 3 to generate the "boss's" logical reaction.
- **Text-to-Speech (TTS):** Sends the AI response to the ElevenLabs API to generate ultra-realistic, emotionally weighted audio, which is then played back to the user via Vue.

#### The Importance:

- **Real-World Impact:** Practicing difficult conversations out loud builds muscle memory. The realistic audio prevents it from feeling like a sterile video game, triggering real emotional regulation practice.
- **Hackathon Strategy:** A lock for the

*MLH*

Best Use of ElevenLabs prize. The "Walkie-Talkie" turn-based architecture ensures a flawless, latency-free demo without the nightmare of real-time WebRTC bugs.

### 4. Facial Gym (Computer Vision)

#### Functionality:

- **Static Reading:** A multiple-choice quiz where users decode complex facial expressions from static images.
- **Dynamic Masking:** Uses `face-api.js` running locally in the browser to overlay a webcam feed. It tracks emotional arrays (happy, sad, neutral) in real-time while the user is tasked with maintaining a specific expression (e.g., "Stay neutral while listening to this annoying story").

### The Importance:

- **Real-World Impact:** Immensely valuable for neurodivergent users (or anyone with social anxiety) who struggle with reading physical cues or managing their own resting facial expressions.
- **Hackathon Strategy:** Major points for **Accessibility** and **Best Jr Hack**. Running computer vision entirely client-side proves you understand browser constraints and advanced JavaScript integration beyond standard web forms.

## 5. The Infrastructure (Vultr + Local LLM)

### Functionality:

- **Custom Cloud Hosting:** A Vultr GPU instance running Ollama to serve Llama 3 locally.
- **API Gateway:** A Python Flask backend that safely handles API keys (ElevenLabs, Mongo) and routes requests from the Vue frontend to the local LLM.

### The Importance:

- **Real-World Impact:** Guarantees absolute data privacy. Users input highly sensitive, vulnerable social interactions; running an open-source model locally ensures their data isn't being scraped by big tech training algorithms.
- **Hackathon Strategy:** Targets the

*MLH*

Best Use of Vultr prize. Explaining that you self-hosted an AI model because you are under 18 and couldn't use standard API keys is the ultimate hacker flex. It shows resilience, problem-solving, and deep technical learning.e